

Test Report Data Processing on Modern Hardware Assignment5

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Test on LxHalle

perf data in repo

Lock is very expensive

- **lockCoupling**
 - Highest time, CPU cycles, and instruction counts.
 - High cache misses (especially L1 and LLC) → frequent cache line invalidation due to fine-grained locking.
 - Very low IPC → threads spend most of their time waiting for locks.
- **optimisticLockCoupling**

- Slightly better than `lockCoupling`, but still expensive.
- Higher IPC than `lockCoupling` but lower than optimistic or nosync methods.
- Still suffers from lock contention and retry overhead.
- `optimistic` achieve best performance



